Web Technologies

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What is HTML?

HTML = HyperText Markup Language

Standard markup language for structuring Web content

HTML describes what content is: headings, paragraphs, links, images, tables, media

HTML is a markup language, not a programming language

Markup language = text + special tags/notation that describe how the text is structured or presented.

What is Markup?

Markup adds information about the structure or meaning of content

Example: <h1> identifies a heading; <p> identifies a paragraph

HTML focuses on content structure and meaning rather than presentation alone

Why Do We Need HTML?

Provides structure to webpage content

Helps browsers understand content

Supports accessibility and screen readers

Helps search engines understand page structure

Connects documents and embeds resources

HTML vs CSS vs JavaScript

HTML → structure

CSS → presentation and visual design

JavaScript → behavior and interactivity

Analogy: HTML = skeleton, CSS = appearance, JavaScript = behavior

Basic HTML Document

<!DOCTYPE html>

<html>

<head> … </head>

<body> … </body>

The browser interprets this structure and renders the webpage

DOCTYPE

<!DOCTYPE html> declares a modern HTML document

Written at the beginning of the document

It is a declaration, not an HTML element

Helps browsers use standards mode

HTML5 uses the simple DOCTYPE shown above

The <html> Element

Root element of the HTML document

All document content except DOCTYPE is inside <html>

Use the lang attribute to identify the document language

Example: <html lang="en">

Language information supports accessibility and search engines

The <head> Section

Contains metadata and information about the document

Common elements: <title>, <meta>, <link>, <style>, <script>

Most head content is not directly visible in the webpage

Example: <meta charset="UTF-8"> and <title>My Page</title>

The <body> Section

Contains the visible/main webpage content

May include headings, paragraphs, images, links, lists, tables, forms and media

Example: <h1>Welcome</h1> and My first webpage.

Complete HTML5 Structure

DOCTYPE

<html lang="en">

<head> → metadata, title

<body> → visible content

Think of the document as a tree: html → head + body

Elements and Tags

Tag: markup inside angle brackets, e.g.

Element: opening tag + content + closing tag

Example: Hello World

Opening tag:

Closing tag:

Void Elements

Some HTML elements do not contain content and do not require closing tags

Examples:
, <hr>, <img>, <meta>, <link>, <input>

HTML5 commonly uses
, not XHTML-style

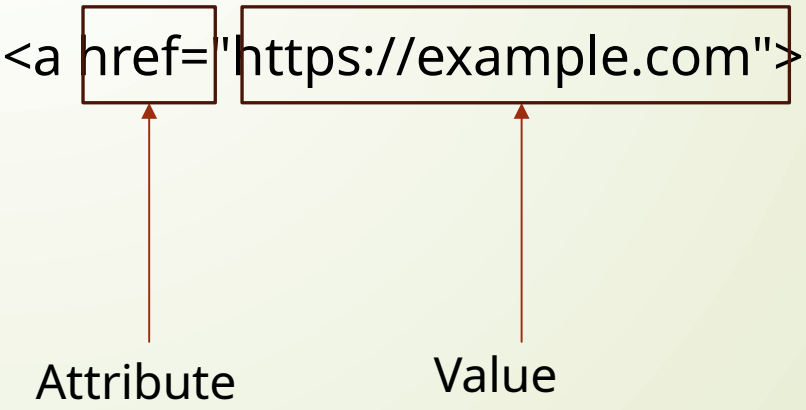
Attributes

Attributes provide additional information about an element

General form: <tag attribute="value">

Examples: href, src, alt, id, class, title, width, height

Example: Example



HTML Terminology

- Concepts in HTML
  - Tags
    - Opening tag and closing tag
    - The smallest piece in HTML
  - Attributes
    - Properties of the tag
    - Size, color, etc…
  - Elements
    - Combination of opening, closing tag and attributes

HTML Elements

- HTML Elements are combination of tags and attributes
  - Opening tag with some or none attributes and a closing tag

```
<a href="http://google.com"> go to Google</a>
```

```
<html>…</html>
```

Common HTML Attributes

id → identifies an element

class → groups elements

style → inline CSS

title → additional information

href → link destination

src → resource location

alt → alternative image text

Nesting Elements

HTML elements can be placed inside other elements

Correct: This is important.

Elements should be properly nested

Incorrect: Hello

Think of nested elements as a parent-child tree

HTML Comments

Comments are notes for developers

Syntax: <!-- This is a comment -->

Comments are not displayed on the webpage

Useful for documentation and temporarily disabling code

HTML Validation

Validation checks whether HTML follows the rules of the standard

Can identify structural and syntax problems

Examples: incorrect nesting, invalid attributes, missing required information

Benefits: better code quality, standards compliance and cross-browser reliability

HTML Validator

- An HTML validator is a specialized program or application used to check the validity of HTML markup in a Web page for any syntax and lexical errors.
- This is because HTML is not compiled and it is a very forgiving language in terms of errors; everything still works in a way, so there is no quick way to check for such errors.
- Hence, a separate program is required.
- The W3C HTML Validator is a way to validate your HTML
- http://validator.w3.org/

Headings

HTML provides six heading levels: <h1> through <h6>

Use headings according to document hierarchy

Typical structure: h1 → h2 → h3

Do not choose headings only because they look larger; CSS controls appearance

Paragraphs and Text Formatting

<p> → paragraph

 → strong importance

 → emphasis

<mark> → highlighted text

<small> → smaller side text

<del> → deleted text

<ins> → inserted text

<sub> and <sup> → subscript and superscript

Text Formatting

- Text formatting tags modify the text between the opening tag and the closing tag
  - Ex. Hello makes "Hello" bold

	bold
	italicized
<u></u>	underlined
	Sample^{superscript}
	Sample$_{subscript}$
	strong
	emphasized
<pre></pre>	`Preformatted text`

- **Many of the formatting tags are deprecated**
  - **Use CSS instead**

Hyperlinks

The <a> element creates hyperlinks

Example: Google

href specifies the destination

Links can point to webpages, files, sections, email addresses and other resources

Absolute vs Relative URLs

Absolute URL: complete address, e.g. https://www.comsats.edu.pk

Relative URL: location relative to the current document, e.g. about.html

Use absolute URLs commonly for external websites

Use relative URLs commonly for pages/files within the same website

Images

Use <img> to display an image

Example: <img src="campus.jpg" alt="University campus">

src → image location

alt → alternative text

Width/height can control dimensions

Meaningful alt text improves accessibility

Audio and Video

HTML5 provides native media elements

Audio: <audio controls> … </audio>

Video: <video controls> … </video>

The <source> element can specify media files and MIME types

controls provides browser playback controls

Embedding External Content with <iframe>

<iframe> embeds external content in a webpage

Common uses: YouTube videos, maps and other embeddable content

Example: <iframe src="https://www.youtube.com/embed/VIDEO_ID" title="Video"></iframe>

Use an appropriate width, height and descriptive title

HTML Lists

Unordered list: <ul> with <li> items

Ordered list: <ol> with <li> items

Example: HTML
CSS

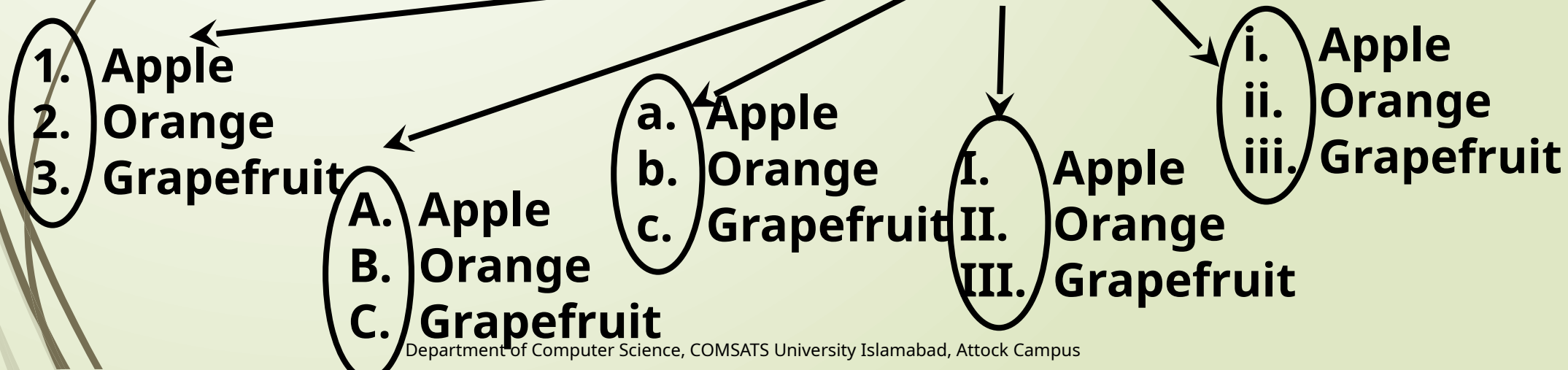
Lists are useful for menus, steps, features and collections

Ordered Lists: <ol> Tag

- Create an Ordered List using :

```
<ol type="1">
  <li>Apple</li>
  <li>Orange</li>
  <li>Grapefruit</li>
</ol>
```

- Attribute values for type are 1, A, a, I, or i

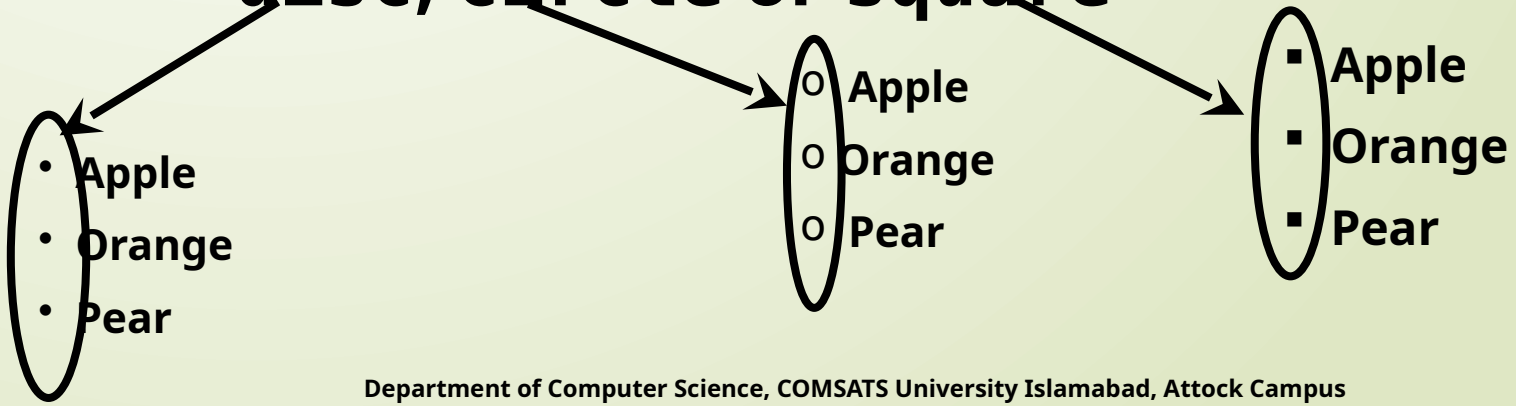


Unordered Lists: <ul> Tag

- Create an Unordered List using :

```
<ul type="disc">
  <li>Apple</li>
  <li>Orange</li>
  <li>Grapefruit</li>
</ul>
```

- Attribute values for type are:
  - disc, circle or square



Definition lists: <dl> tag

- Create definition lists using <dl>
  - Pairs of text and associated definition; text is in <dt> tag, definition in <dd> tag

```
<dl>
   <dt>HTML</dt>
   <dd>A markup language …</dd>
   <dt>CSS</dt>
   <dd>Language used to …</dd>
</dl>
```

Output:

HTML
    A markup language …
CSS
    Language used to …

  - Renders without bullets
  - Definition is indented

HTML Tables

Tables organize data into rows and columns

<table> → table

<tr> → row

<th> → header cell

<td> → data cell

Use tables for tabular data, not page layout

HTML Tables

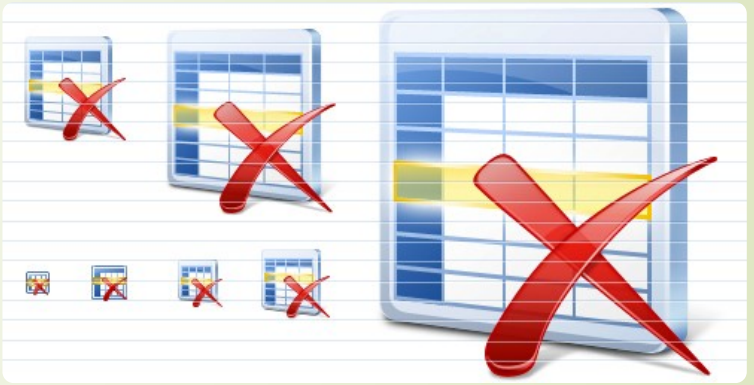
- Tables represent tabular data
  - A table consists of one or several rows
  - Each row has one or more columns
- Tables are comprised of several core tags:
  - `<table></table>`: begin/end table definition
  - `<tr></tr>`: create a table row
  - `<td></td>`: create tabular data (cell)
- Tables should not be used for layout
  - Use CSS floats and positioning styles instead

Simple HTML Tables – Example

```
<table cellspacing="0" cellpadding="5">
  <tr>
    <td><img src="ppt.gif"></td>
    <td><a href="lecture1.ppt">Lecture
1</a></td>
  </tr>
  <tr>
    <td><img src="ppt.gif"></td>
    <td><a href="lecture2.ppt">Lecture
2</a></td>
  </tr>
  <tr>
    <td><img src="zip.gif"></td>
    <td><a href="lecture2-demos.zip">
      Lecture 2 - Demos</a></td>
  </tr>
```

Important Birthdays

Name	Age	Date of Birthday
Emily	8 years	3 September
Jack	12 years	28 April



Simple HTML Tables

Data Cells and Header Cells

- Two kinds of cells in HTML tables
  - Data cells – containing the table data
  - Header cells – used for the column names or some more important cells
- Why two kinds of cells?
  - Used to semantically separate the cells

```
<tr>
    <th>Full Name</th>
    <th>Mark</th>
</tr>
<tr>
    <td>Doncho Minkov</td>
    <td>Very good (5)</td>
</tr>
<tr>
    <td>Georgi Georgiev</td>
    <td>Exellent (6)</td>
</tr>
```

Full Name	Mark
Doncho Minkov	Very good (5)
Georgi Georgiev	Exellent (6)

Complete HTML Tables

- Table rows split into three semantic sections: header, body and footer
  - <thead> denotes table header and contains <th> elements, instead of <td> elements
  - <tbody> denotes a collection of table rows that contain the very data
  - <tfoot> denotes table footer.
  - <colgroup> and <col> define columns (used to set column widths)

Complete HTML Table: Example

```
<table>
  <colgroup>
    <col style="width:100px" /><col />
  </colgroup>
  <thead>
    <tr><th>Column 1</th><th>Column
  2</th></tr>
  </thead>
  <tbody>
    <tr><td>Cell 1
  1.2</td></tr>
    <tr><td>Cell 2
  2.2</td></tr>
  </tbody>
  <tfoot>
    <tr><td>Footer 1</td><td>Foote
```

columns

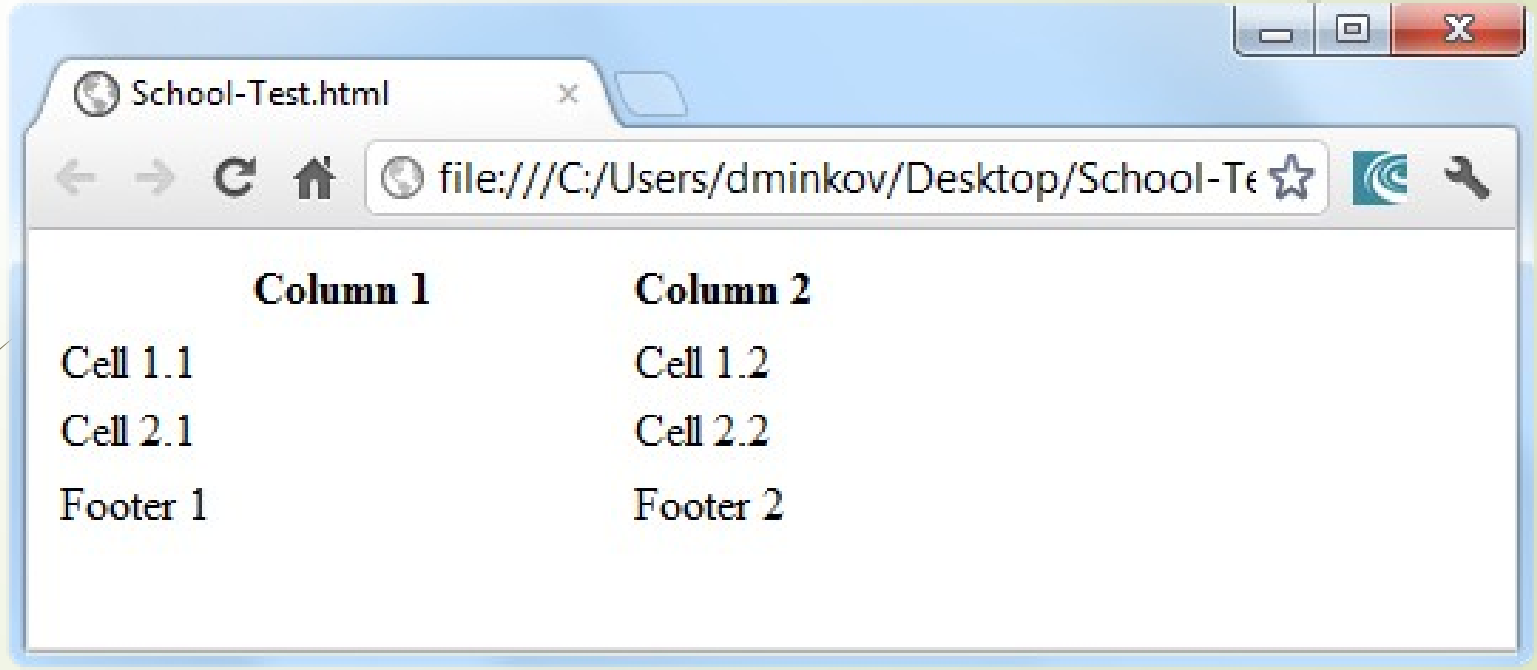
header

th

footer

Last comes the body (data)

Complete HTML Table:Example





Nested HTML Tables

Nested Tables

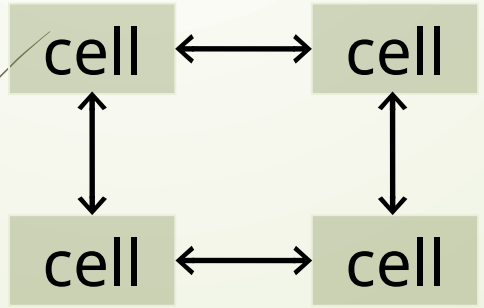
- Table "cells" (<td>) can contain nested tables (tables within tables):

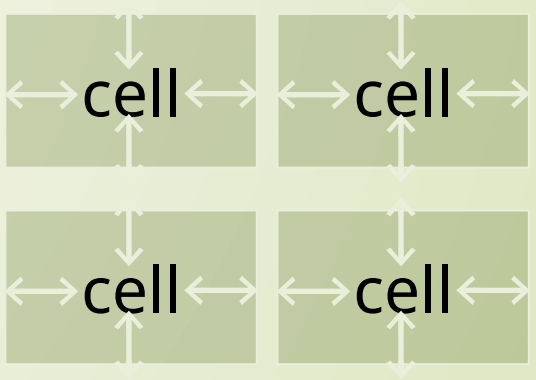
```
<table>
  <tr>
    <td>Contact:</td>
    <td>
      <table>
        <tr>
          <td>First Name</td>
          <td>Last Name</td>
        </tr>
      </table>
    </td>
  </tr>
</table>
```



Cell Spacing and Padding

- Tables have two attributes related to space
  - cellspacing
  - cellpadding





- Defines the empty space between cells
- Defines the empty space around the cell content



Row and Column Spans

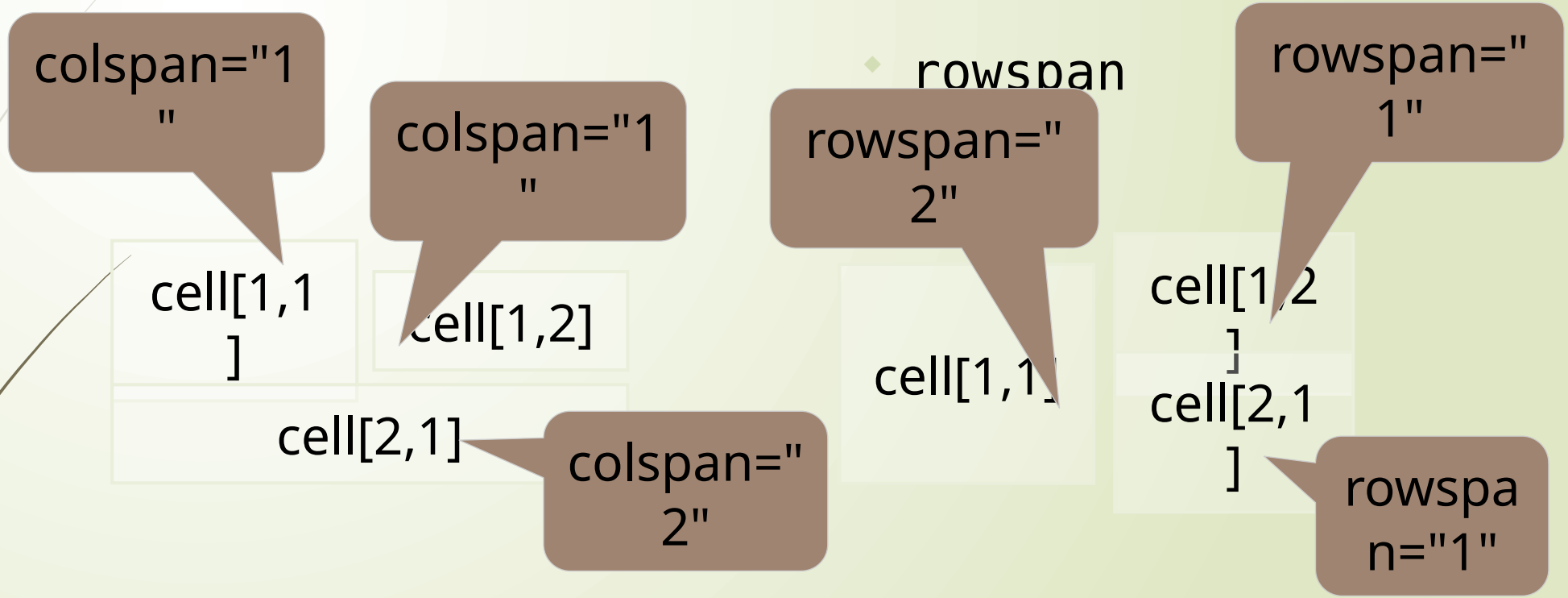
How to Make a Two-Cells Column or Row?





Column and Row Span

- Cells have two attributes related to merging



- Defines how many columns the cell occupies
- Defines how many rows the cell occupies

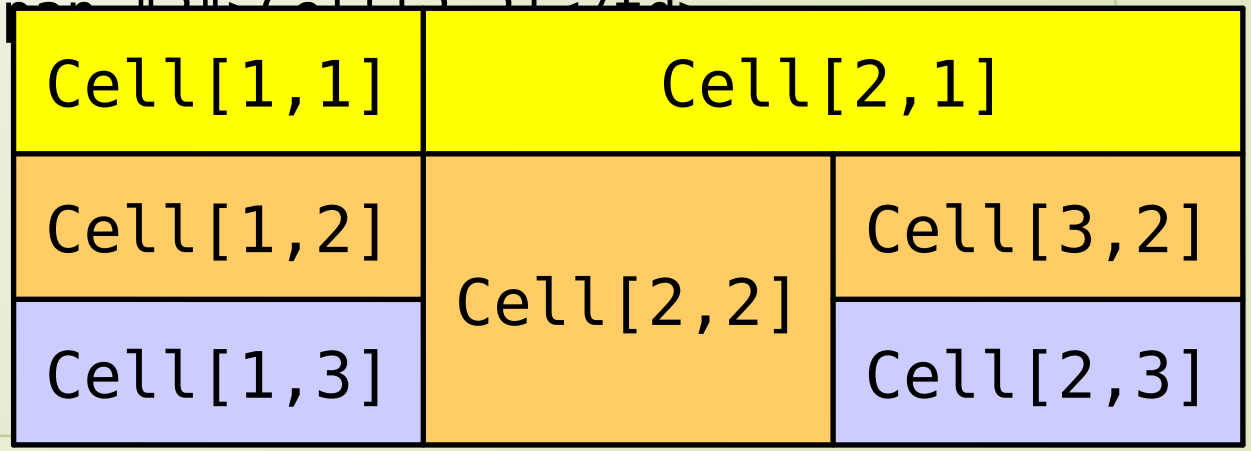
Column and Row Span

```
<table cellspacing="0">
   <tr class="1">
         <td>Cell[1,1]</td>
         <td colspan="2">Cell[2,1]</td>
   </tr>
   <tr class="2">
         <td>Cell[1,2]</td>
         <td rowspan="2">Cell[2,2]</td>
         <td>Cell[3,2]</td></tr>
   <tr class="3">
         <td>Cell[1,3]</td>
         <td>Cell[2,3]</td><
   /tr>
</table>
```

Column and Row Span

table-colspan-rowspan.html

```
<table cellspacing="0">
   <tr class="1">
         <td>Cell[1,1]</td>
         <td colspan="2">Cell[2,1]</td>
   </tr>
   <tr class="2">
         <td>Cell[1,2]</td>
         <td rowsp
         <td>Cell
   <tr class="3">
         <td>Cell
         <td>Cell
   /tr>
</table>
```



What is HTML5?

HTML5 is the modern major version of HTML

Introduced richer semantic structure

Added native audio and video

Improved forms and input types

Provides APIs and capabilities for modern web applications

Examples: <header>, <nav>, <main>, <section>, <article>, <aside>, <footer>

HTML5 Semantic Elements

Semantic elements communicate the role/meaning of content

<header> → introductory/header content

<nav> → navigation

<main> → main content

<section> → thematic section

<article> → self-contained content

<aside> → related content

<footer> → footer information

Benefits: accessibility, maintainability and search-engine understanding

Semantic HTML and Accessibility

Accessibility means making websites usable by people with different abilities and assistive technologies.
Semantic HTML helps assistive technologies understand page structure.

```
<nav>
   ...
</nav>
```

```
<article>
   <h1>Introduction to Artificial Intelligence</h1>
   <p>Artificial intelligence...</p>
</article>
```

HTML5 Attributes, Browser Support & Summary

New/useful form attributes and input types include required, placeholder, autofocus, multiple, email, date and number

Modern Chrome, Edge, Firefox and Safari support most commonly used HTML5 features

Support can vary for newer or specialized APIs, so check compatibility when needed

Key idea: HTML provides structure; HTML5 adds semantic, multimedia, form and web-app capabilities

Practical task: create a page with headings, paragraphs, links, image, lists, table, media and a semantic element

Block Level Elements

- A block-level element always starts on a new line, and the browsers automatically add some space (a margin) before and after the element.
- A block-level element always takes up the full width available (stretches out to the left and right as far as it can).
- Two commonly used block elements are: <p> and .

- https://www.w3schools.com/html/html_blocks.asp

Inline Elements

- An inline element does not start on a new line.
- An inline element only takes up as much width as necessary.
- This is a element inside a paragraph.

- https://www.w3schools.com/html/html_blocks.asp

The Tag

- creates logical divisions within a page
  - Block element
  - Used with CSS
- Example:




```
<div style="font-size:24px; color:red">DIV example</div>

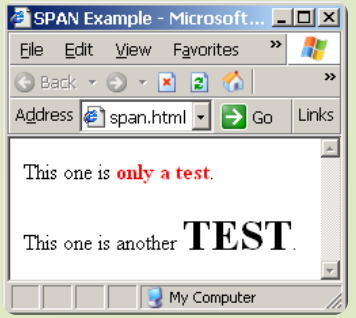
<p>This one is <span style="color:red; font-weight:bold">only a test.</span></p>
```

The Tag

- Inline style element
- Useful for modifying a specific portion of text
  - Don't create a separate area (paragraph) in the document
- Mainly used to style parts of a text

```
<p>This one is <span style="color:red; font-
weight:bold">only a test.</span></p>

<p>This one is another <span style="font-
size:32px; font-weight:bold">TEST</span>.</p>
```



HTML Forms

HTML Forms

- Forms are the primary method for gathering data from site visitors.
- Create a form block with

```
<form></form>
```

- Example:

```
<form id="myForm" method="post"
action="path/to/some-script.php" >
        ....
</form>
```

The “method" attribute tells how the form data should be sent – via GET or POST request

The "action" attribute tells where the form data should be sent

Common Attributes

Type	**what kind of input field** it should display.
Name	The field's name/key when sending data.
Id	unique **identifier for the element on the webpage**.
Value	actual value associated with the input
Placeholder	displays a temporary **hint/example**

```
<label for="username">Username:</label>

<input
   type="text"
   id="username"
   name="username"
   value=""
   placeholder="Enter username"
   required
   pattern="[A-Za-z]{3,}">
```

Client-Side Validation Attributes

- Required — user **must provide a value** before submitting the form.
- Min — **minimum allowed value** for applicable input types
- Max — max specifies the **maximum allowed value**.
- Pattern — specifies a **regular expression (regex)** that the input must match.

Form Fields

- Text fields are single-line entry fields:

```
<input type="text" name="FirstName"
value="This is a text field">
```

- Text areas can contain multiple lines of text:

```
<textarea name="Comments">This is a multi-
line text field</textarea>
```

- Hidden fields contain data not shown to user:

```
<input type="hidden" name="Account"
value="This is a hidden text field">
```

Form Input Controls

- Create a checkbox:

```
<input type="checkbox" name="fruit"
value="apple">
```

- Create a radio button:

```
<input type="radio" name="title"
value="Mr.">
```

- Radio buttons can be grouped, allowing only one to be selected from a group:

```
<input type="radio" name="town"
value="Sofia">
<input type="radio" name="town"
value="Varna">
```

Other Form Controls

- Pull down menu (drop-down list):

```
<select name="gender">
  <option value="Value 1"
    selected="selected">Male</option>
  <option value="Value 2">Female</option>
  <option value="Value 3">Other</option>
</select>
```

- Submit button:

```
<button type="submit">Submit</button>
```

Other Form Controls (2)

- Reset button – clears the form

```
<button type="reset">Reset
Button</button>
```

- Image button – acts like submit but image is displayed instead of button

```
<input type="image" src=“image.jpg"
alt="Submit" width="100" height="50">
```

- Ordinary button – used for JavaScript, no default action

```
<input type="button" value="simple
button">
```

Other Form Controls (3)

- Password input – acts like normal text field but hides the text with * signs

```
<input type="password" name="pass"
value="">
```

- Multiple select field – code is like drop down but displays list of items to select

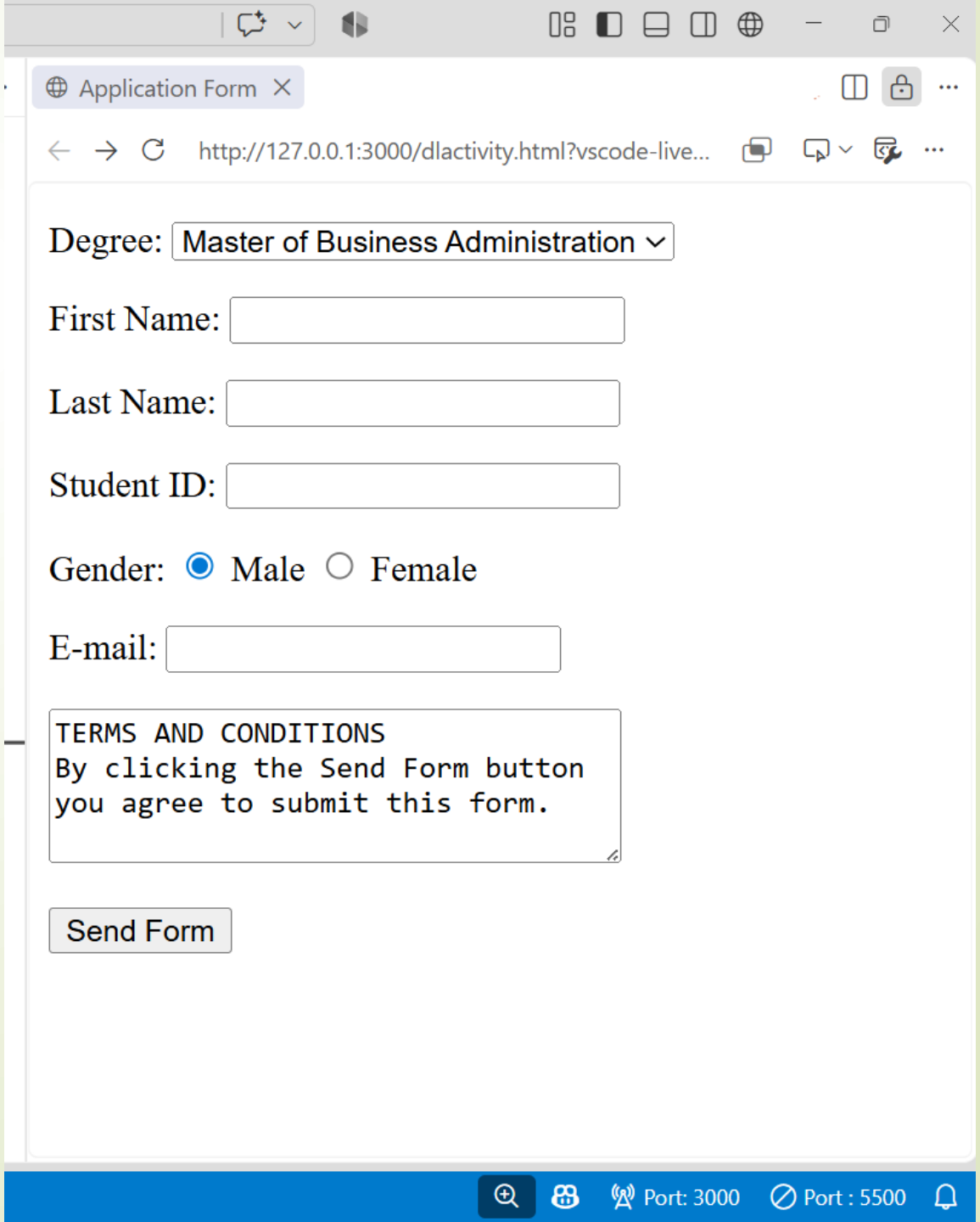
```
<select name="products"
multiple="multiple">
  <option value="Value 1"
    selected="selected">keyboard</option>
  <option value="Value 2">mouse</option>
  <option value="Value
3">speakers</option>
```

HTML Forms – Example

```
<form method="POST" action="apply-now.php">
  <input name="subject" type="hidden" value="Class”>
  <p>Degree:
    <select name="degree">
      <option value="BA">Bachelors of Art</option>
      <option value="BS">Bachelors of Science</option>
      <option value="MBA" selected>Masters of
        Business Administration</option>
    </select><form method="POST" action="apply-now.php">
        <input name="subject" type="hidden" value="Class">
        <p>
            Degree:
            <select name="degree">
                <option value="BA">Bachelor of Arts</option>
                <option value="BS">Bachelor of Science</option>
                <option value="MBA" selected>Master of Business Administration</option>
            </select>
        </p>
        <p>
            First Name:
            <input type="text" name="firstname">
        </p>
        <p>
            Last Name:
            <input type="text" name="lastname">
        </p>
        <p>
            Student ID:
            <input type="password" name="studentid">
        </p>
```

HTML Forms – Example (2)

```
        <p>
            Gender:
            <input name="gender" type="radio" value="Male" checked> Male
            <input name="gender" type="radio" value="Female"> Female
        </p>
        <p>
            E-mail:
            <input type="email" name="email">
        </p>
        <p>
            <textarea name="terms" cols="30" rows="4" readonly>
TERMS AND CONDITIONS
By clicking the Send Form button you agree to submit this form.
        </textarea>
        </p>
        <p>
            <input type="submit" name="button" value="Send Form">
        </p>
    </form>
```


Application Form
http://127.0.0.1:3000/dlactivity.html?vscode-live...
Degree: Master of Business Administration
First Name:
Last Name:
Student ID:
Gender: Male Female
E-mail:
TERMS AND CONDITIONS
By clicking the Send Form button
you agree to submit this form.
Send Form
Port: 3000
Port : 5500